

Lecture 23: Testing of Hypotheses - Preliminary Example

Since we cannot simultaneously decrease ~~the~~ the probs. of ~~both~~ both Type I & Type II error, we fix one and minimize the other in the following manner:

Step 1

Fix, the level of significance ^{of the test}, say, $\alpha \in (0, 1)$, usually, a small fraction α

max Prob. (Type I error) $\leq \alpha$, i.e.,

$$\sup_{\theta \in \Theta_0} P_\theta(X \in W) \leq \alpha \quad \left(\begin{array}{l} \text{'=' can be achieved by continuous distns,} \\ \text{but may not be so for discrete distns } \forall \alpha \in (0, 1) \end{array} \right)$$

$\sup_{\theta \in \Theta_0} P_\theta(X \in W)$ is called the size of the test & satisfying the condition size $\leq$ level, makes it a size- α test (exact size- α test if size = level).

From the above discussion & terminologies used

$$\text{Prob. (Type I error for some } \theta \in \Theta_0) \leq \text{Size of the test} \leq \text{Level of significance of the test.}$$

(For a continuous distn & a singleton Θ_0 , we get '=' throughout.)

~~Step 1~~ Now, define $P_\theta(X \in W | \theta \in \Theta)$ = Power function of the test, say, $P(\theta)$ so that and $P_\theta(X \in W | \theta \in \Theta_0)$ = Power of the test

Step 2 After fixing α , i.e., the max value of the Prob. of Type I error, we minimize Prob. of Type II error, i.e., minimize $1 - \text{Power of the test}$, i.e., maximizing, the Power of the test.

For ease of the math involved, we quantify the decision (accept/reject) through a test fn
$$\phi(x) = \begin{cases} 1 & \text{if } x \in W \\ 0 & \text{if } x \notin W \end{cases}$$

In terms of the test fn ϕ , we define a size- α (or, exact size- α) test as $\sup_{\theta \in \Theta_0} E_\theta \phi(X) \leq (\text{or, } =) \alpha$, and its power function by $P_\phi(\theta) = E_\theta \phi(X), \theta \in \Theta$.

In most situations, the critical region will be expressible in terms of a statistic, i.e., a function of the random observations, rather than through the ~~the~~ such obs_{ns} individually, i.e.,

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$$\{\underline{x} \in \omega\} \Leftrightarrow \{\underline{x} : T(\underline{x}) < c_1\} \quad \text{to give a } \begin{array}{l} \text{left-tailed} \\ \text{right-tailed} \end{array} \text{ test}$$

$$\text{or } \{\underline{x} : T(\underline{x}) > c_2\}$$

$$\text{or } \{\underline{x} : d_1 < T(\underline{x}) \leq d_2\} \quad \text{2-tailed}$$

The test for ϕ , introduced here is a ~~non~~ randomized test. A randomized test will be required if we insist for an exact size- α test even with a discrete dist_n (More later!).

Example 1. $X \sim f_\theta$, $f_\theta(x) = \theta x^{\theta-1}$, $0 < x < 1$; $\theta > 0$
For testing, $H_0: \theta = 1$ ag. $H_A: \theta = 2$ with 2 observations, the following critical region is ~~const~~ suggested:
 $\omega = \{(x_1, x_2) : \frac{1}{2x_1} \leq x_2\}$. Find the size & power of the test (having this c.r.).

Power for $P(\theta) = P\left(\frac{1}{2x_1} \leq x_2\right) = \theta^2 \int_{\frac{1}{2}}^1 \int_{\frac{1}{2x_1}}^1 (x_1 x_2)^{\theta-1} dx_2 dx_1$
 $= 1 - \left(\frac{1}{2}\right)^\theta (1 - \theta \log(3/4)).$

Then, Size of the test (= Prob. of Type I err)
 $= P(\theta = 1) = 0.356$ (This is also the level of the test)

and power of the test
 $= P(\theta = 2) = 0.606$.

Next, the Neyman-Pearson lemma for construction of critical region ω , instead of suggesting ad-hoc critical region as in this example.